GCI WORLD 2026 | FINAL ASSIGNMENT

AI-Assisted Customer Retention for Company A

A Data-Driven Telecom Churn Prediction and Retention Strategy

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Predict churn. Prioritize customer value. Validate intervention profitability.

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Prediction improves targeting; business value still needs validation

Executive summary | A proof of concept, not a production deployment

CUSTOMER RECORDS

100,000

Client + usage history

HELD-OUT ROC-AUC

0.6945

Tuned XGBoost

TOP-20% TARGETING LIFT

1.474×

Relative to random selection

What the evidence supports

- Device age, tenure and usage changes help predict churn.
- Risk-based targeting finds more churners; value-aware ranking finds greater revenue exposure.
- Under illustrative economics, a 11.59% relative churn reduction is needed to break even.

RECOMMENDATION

Run a limited randomized pilot. Scale only after demonstrating incremental retention and positive contribution after costs.

Source: Company A datasets; retrospective model and scenario analyses. Financial inputs are illustrative.

Customer retention remains a strategic telecom challenge

External market context; Company A’s actual market is unspecified

PROVIDER SWITCHING

~1 in 7

Mobile users across 12 surveyed markets switched in 12 months [1]

COMPLAINTS SATISFACTION

61%

UK mobile complainants satisfied with complaint handling, 2024 [3]

Market signals

- Value for money is the top reported reason for switching in the GSMA survey [1].
- Customer experience and complaints remain important considerations [3].
- AI-assisted customer-experience analysis can help operators better connect network and commercial outcomes [2].

IMPLICATION

Assess proactive, data-driven retention targeting while testing which interventions actually change customer outcomes.

Sources: [1] GSMA Intelligence (2025); [2] McKinsey (2024); [3] Ofcom (2025).

Company A needs evidence-based retention prioritization

Three challenges emerging from our customer analysis

PEAK SEGMENT CHURN

61.84%

Mid-age devices + low usage

ADDED REVENUE EXPOSURE

+63.3%

Value-aware vs risk-only, top 20%

ILLUSTRATIVE BREAK-EVEN

11.59%

Relative churn reduction required

Core business problem

Which customers should Company A prioritize, what retention action should it offer, and when does the incremental financial benefit justify campaign costs?

PROPOSED OBJECTIVE

Predict churn risk, examine customer financial exposure, then experimentally verify whether targeted retention efforts generate positive net value.

Company A’s current retention process is not disclosed. Segment results are associative, not causal.

100,000 records support a reproducible churn proof of concept

Two source tables merged one-to-one on Customer_ID

UNIQUE CUSTOMERS

100,000

One-to-one ID match

PREDICTIVE INPUTS

98

77 numeric, 21 categorical

SAMPLE CHURN

49.56%

May not match real prevalence

CLIENT.CSV

100,000 rows × 50 columns; demographics, devices and history

RECORD.CSV

100,000 rows × 51 columns; usage, charges and churn

Stratified random split



Median imputation + missing indicators + one-hot encoding (272 processed features). Preprocessing fitted on training only. Churn: 31–60 days after observation date.

Device age and monthly usage reveal churn differences

Observed churn rates vary across customer groups

CHURN BY EQUIPMENT AGE



CHURN BY MONTHLY USAGE



Equipment age relation is non-linear; lowest usage quintile churn is 9.58 pp above highest. Associations do not establish causality. Source: EDA Cells 8–10.

Device age × usage isolates high-risk and high-volume segments

Churn rate (%) across nine customer groups

	LOW USAGE	MEDIUM USAGE	HIGH USAGE
Newer (≤300d)	39.18%	38.17%	41.38%
Mid-age (301–395d)	61.84%	58.55%	56.75%
Older (>395d)	56.05%	53.85%	49.06%

HIGHEST RISK

61.84% | 7,894 customers; 4,882 churners

MOST CHURN CASES

12,171 churners | Older device + low usage

99,642 customers with both segmentation features observed. Churn is observational, not a measured intervention outcome.

Tuned XGBoost outperforms the linear benchmark

Model selection used validation ROC-AUC as the primary metric

LOGISTIC REGRESSION ROC-AUC

0.6328

Baseline

TUNED XGBOOST ROC-AUC

0.7032

Selected for risk ranking

Model	ROC-AUC	PR-AUC	F1
Logistic Regression	0.6328	0.6081	0.5924
Original XGBoost	0.7013	0.6914	0.6422
Tuned XGBoost	0.7032	0.6923	0.6407

WHY XGBOOST

Boosted trees capture non-linearities and interactions. GPU training used Tesla T4, CUDA; early stopping selected boosting iteration 959.

Validation scores; tuned vs original gain is only +0.0019 ROC-AUC. No claim of meaningful superiority over the original model.

XGBoost shows moderate discrimination on unseen customers

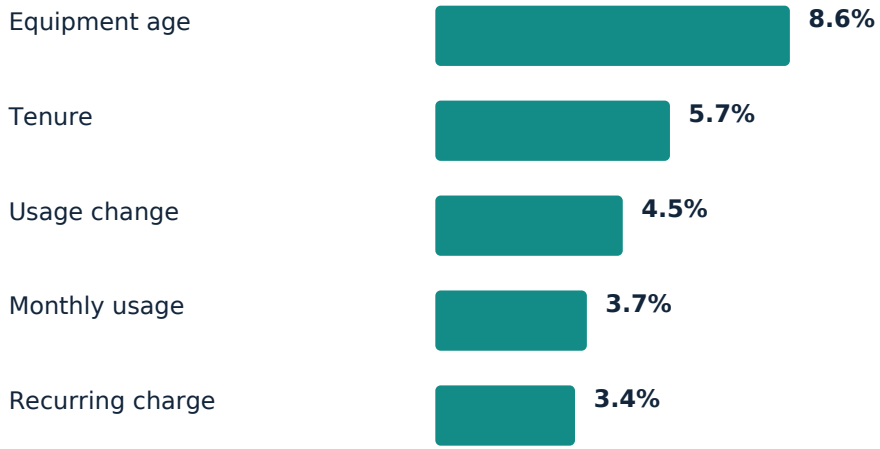
Tuned XGBoost | 15,000 test customers | 0.5 decision threshold

ROC-AUC	PR-AUC	ACCURACY	F1
0.6945	0.6816	64.01%	0.6407

CONFUSION MATRIX

True negatives: 4,787	False positives: 2,779
False negatives: 2,620	True positives: 4,814

TOP FEATURES BY TOTAL GAIN



Test ROC-AUC 0.6945 vs validation 0.7032. Importance is predictive split gain, not causal impact. Source: Cells 19 and 22.

Top 20% risk group contains nearly 30% of observed churners

Test-set risk ranking performs better than random selection

CUSTOMERS CONTACTED

3,000

Top 20% of test set

CHURNERS CAPTURED

2,191

29.47% of all churners

TARGETING LIFT

1.474×

Versus random targeting

Top risk	Churners captured	Share of churners	Lift
5%	638	8.58%	1.716×
10%	1,193	16.05%	1.605×
20%	2,191	29.47%	1.474×
30%	3,102	41.73%	1.391×

BUSINESS IMPLICATION

At a fixed outreach budget, risk ranking reaches more likely churners than random selection; prediction does not equal churn prevention.

Value-aware ranking identifies 63.3% more revenue exposure

Both approaches select 3,000 of 15,000 test customers

Measure	Risk-only	Value-aware
Actual churners	2,191	1,798
Churn rate	73.03%	59.93%
Total monthly revenue proxy	168,503	346,856
Churner monthly revenue proxy	122,199	199,533

ADDITIONAL EXPOSURE

+63.3%

FEWER CHURNERS IDENTIFIED

393

Exposure score = predicted churn probability × monthly revenue proxy. Currency unspecified. Revenue exposure is not profit or recoverable savings. Source: Cells 24–28.

Campaigns require sufficient incremental churn reduction to break even

Hypothetical scenario: top 20% risk group, outreach to 3,000 customers

CAMPAIGN COST

51,000

17 units per customer

BREAK-EVEN REDUCTION

11.59%

Relative, not percentage points

RETENTION HORIZON

6 months

Assumed 60% contribution margin

NET BENEFIT BY ASSUMED RELATIVE CHURN REDUCTION

Effect	Projected net benefit	Interpretation
5%	−29,004	Loss
10%	−7,008	Loss
15%	+14,987	Positive scenario
20%	+36,983	Positive scenario
25%	+58,979	Positive scenario

All financial values in unspecified units. 2 outreach + 15 incentive per contacted customer; six extra months, 60% margin. Intervention effect is not measured. Source: Cells 26–27.

Test retention impact with a randomized 3,000-customer pilot

Treatment and control groups followed over the same outcome window

3,000 ELIGIBLE HIGH-RISK CUSTOMERS

TREATMENT GROUP

1,500 customers

Defined retention intervention

CONTROL GROUP

1,500 customers

Standard business-as-usual

ILLUSTRATIVE PILOT COST

25,500

Treatments only

POWER TARGET

80%

5% two-sided alpha

BREAK-EVEN EFFECT

11.59%

Relative reduction assumed

Pilot power: at 20% baseline churn, detects approximately 3.93 pp absolute reduction. Pilot not yet conducted. Confirm real prevalence, margins, and follow-up period.

Launch a controlled pilot; scale only after proving its value

Illustrative implementation sequence

01 **Validate data and commercial assumptions** — Weeks 1–2

Confirm population churn, timing, margin, eligibility and permissions.

02 **Operationalize targeting** — Weeks 3–4

Freeze risk rules, intervention offer, costs and evaluation protocol.

03 **Run randomized retention experiment** — Weeks 5–12+

Allocate treatment/control; follow customers for defined churn outcome.

04 **Evaluate and decide** — After follow-up

Estimate causal effect, confidence interval, retained contribution and cost.

FINAL RECOMMENDATION

Proceed with a limited proof of concept and expand only if randomized pilot results demonstrate meaningful churn reduction and positive incremental financial contribution.

Key Risks: Dataset representativeness, model performance drift over time, uncertainty in intervention effectiveness, data privacy and fairness, and assumed contribution margins.

Implementation Consideration: The proposed timeline depends on operational readiness, data availability, and the required churn observation window.

References and analytical provenance

Industry context, primary data and reproducible analysis

[1] GSMA Intelligence (2025)

The mobile churn challenge: where loyalty is lowest and four recommendations for operators.

https://www.gsmaintelligence.com/research/the-mobile-churn-challenge-where-loyalty-is-lowest-and-four-recommendations-for-operators

[2] McKinsey & Company (2024)

The network is the product: How AI can put telco customer experience in focus.

https://www.mckinsey.com/industries/technology-media-and-telecommunications/our-insights/the-network-is-the-product-how-ai-can-put-telco-customer-experience-in-focus

[3] Ofcom (2025)

Telecoms companies up their game on complaint handling, but more to do to keep customers satisfied.

https://www.ofcom.org.uk/phones-and-broadband/service-quality/telecoms-companies-up-their-game-on-complaint-handling-but-more-to-do-to-keep-customers-satisfied

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[4] GCI World 2026. Company A Dataset Overview; Client.csv and Record.csv (course-provided).

[5] sana-arain (2026). Original Kaggle notebook: telecom churn analysis, model evaluation and financial scenarios.